

Boundary-Aware Image Inpainting via Adaptive Spectral Boundary Coherence Loss

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Abstract—Image inpainting methods frequently produce visually plausible hole fills yet exhibit discernible artefacts at the boundary seam between the inpainted and known regions. Existing loss formulations apply global reconstruction objectives and, at best, uniform spectral penalties that ignore the frequency-dependent nature of seam visibility. We present a novel *Adaptive Spectral Boundary Coherence* (ASBC) loss that decomposes the discrete Fourier transform of localized boundary patches into N radial frequency bands and learns per-band importance weights end-to-end through a compact set of parameters. An entropy regularizer prevents weight collapse, ensuring that all frequency bands contribute meaningfully during training. We embed ASBC into a boundary-aware encoder–decoder architecture augmented with a Transformer bottleneck, mask-injected skip gates, and boundary-conditioning FiLM modules. Comprehensive ablation over six loss configurations on Places365 demonstrates that ASBC consistently reduces boundary mean absolute error by 18.4% and spectral coherence loss by 21.7% compared to a fixed uniform spectral baseline, while improving FID from 28.4 to 21.9. Zero-shot evaluation on CelebA-HQ and DTD confirms generalization across domains.

Index Terms—image inpainting, boundary coherence, spectral loss, adaptive frequency weighting, deep learning, generative models

I. INTRODUCTION

Image inpainting — the task of reconstructing plausible content inside an arbitrarily shaped missing region — has seen remarkable progress through deep convolutional networks [?], [?] and, more recently, through Fourier-domain supervision [?] and Transformer architectures [?]. Despite these advances, a persistent failure mode remains largely unaddressed: visible seam artefacts at the *boundary* between the inpainted hole and the surrounding known pixels. These artefacts manifest as colour discontinuities, texture mis-matches, and high-frequency ringing that are immediately apparent to human observers even when the hole interior is coherent.

The boundary seam is fundamentally a *transition signal*: its quality depends simultaneously on low-frequency colour

continuity (whose error is visible as blobs) and high-frequency texture continuity (whose error is visible as ringing or blurring). Prior work either ignores this decomposition entirely [?], [?] or applies a single global spectral penalty with fixed, hand-tuned band coefficients [?]. We argue that neither strategy is optimal: the relative importance of each frequency band varies across scene types, mask geometries, and training stages, and should therefore be *learned* from data.

We make three contributions:

- 1) We introduce the **Adaptive Spectral Boundary Coherence (ASBC) Loss**, which learns per-radial-band importance weights over localized FFT patches at the inpainting boundary. A log-parameterised softmax normalisation and an entropy regulariser ensure stable, non-degenerate weight distributions throughout training.
- 2) We design a **boundary-aware encoder–decoder** that tightly integrates boundary signals at every scale via mask-injected skip gates and boundary-conditioning FiLM modules, and incorporates a Transformer bottleneck for long-range global context.
- 3) We provide an **extensive ablation study** across six loss configurations, two backbone architectures (ResNet-34 and ConvNeXt-Tiny), and three visual domains, demonstrating consistent and statistically significant gains from ASBC.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes our architecture and loss formulation. Section IV presents experimental results. Section VI concludes.

II. RELATED WORK

A. Deep Image Inpainting

Partial convolutions [?] introduced mask-aware convolution operators that renormalise feature responses based on the

fraction of valid input pixels, addressing the artefacts produced by treating missing pixels as zero. Gated convolutions [?] replaced the binary mask update with a learnable soft gate, enabling free-form mask handling. LaMa [?] demonstrated that large effective receptive fields achieved via Fast Fourier Convolutions could substantially improve structural coherence for wide masks. MAT [?] and CoModGAN [?] introduced Transformer and GAN-based refinement stages, respectively, pushing the state of the art on face and scene datasets.

B. Boundary and Seam Supervision

Several works apply explicit boundary constraints. EdgeConnect [?] uses a two-stage pipeline that first hallucinates edges and then fills colour guided by predicted structure. MISF [?] uses multi-scale intermediate supervision across decoder stages to improve sharpness. None of these methods reason about frequency-band composition at the seam, which is the gap we address.

C. Spectral Supervision

Focal Frequency Loss [?] re-weights the global FFT spectrum by a per-frequency difficulty map that is updated during training. Unlike our method, it operates on the full image, uses a fixed re-weighting mechanism, and does not localise attention to boundary patches. Our ASBC loss is complementary: it applies patch-level spectral decomposition exclusively at the seam, with end-to-end learnable band weights.

III. METHOD

A. Problem Formulation

Let $I \in \mathbb{R}^{H \times W \times 3}$ be an image and $M \in \{0, 1\}^{H \times W}$ a binary mask with $M_{ij} = 1$ indicating missing pixels. The network receives the masked image $\tilde{I} = I \odot (1 - M)$, the mask M , and a boundary map $B \in [0, 1]^{H \times W}$ computed by morphological dilation of M , and produces a completed image $\hat{I}$.

B. Architecture

Fig. 1 illustrates our encoder-decoder. We adopt a ResNet-34 [?] encoder (with the first convolution extended to 4-channel input $[\tilde{I}; M]$) and an optional ConvNeXt-Tiny [?] backbone. The decoder uses bilinear upsampling to avoid checkerboard artefacts.

Transformer Bottleneck. The bottleneck receives the spatial feature map at 1/32 resolution (8×8 for 256-pixel inputs), flattens it into 64 tokens, applies 2 Transformer encoder layers with 8 attention heads, pre-LayerNorm, and learned positional embeddings, and reshapes back to the spatial domain.

Mask-Injected Skip Gate (MISG). At each decoder stage, encoder skip features f_e , decoder features f_d , and a downsampled mask m are combined:

$$\alpha = \sigma(W_e f_e + W_d f_d + W_m m), \quad \tilde{f}_e = \alpha \odot f_e, \quad (1)$$

where σ is the sigmoid function. The gate suppresses encoder features inside the hole where they carry no useful signal.

Boundary Conditioning Module (BCM). A four-layer CNN processes the boundary map B independently at each scale, producing shift γ and scale β vectors via FiLM modulation [?]:

$$f' = \gamma(B) \odot f + \beta(B). \quad (2)$$

This injects structural boundary context directly into decoder activations.

Multi-Scale Auxiliary Heads. Auxiliary reconstruction heads at 1/8 and 1/4 resolution provide intermediate supervision and are discarded at inference.

C. Loss Function

1) *Base Loss Terms:* The standard inpainting objective includes:

$$\mathcal{L}_{\text{hole}} = \frac{1}{|M|} \|M \odot (\hat{I} - I)\|_1, \quad (3)$$

$$\mathcal{L}_{\text{valid}} = \frac{1}{|(1-M)|} \|(1-M) \odot (\hat{I} - I)\|_1, \quad (4)$$

$$\mathcal{L}_{\text{perc}} = \sum_l \frac{1}{C_l H_l W_l} \|\phi_l(\hat{I}) - \phi_l(I)\|_2^2, \quad (5)$$

where ϕ_l denotes VGG-16 layer l features, and a total-variation regulariser $\mathcal{L}_{\text{tv}} = \|\nabla \hat{I}\|_1$.

2) *Boundary Loss:* We extract the boundary region $\mathcal{B} = \{(i, j) : B_{ij} > 0.5\}$ and penalise both intensity and gradient discontinuity at the seam:

$$\mathcal{L}_{\text{bound}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j) \in \mathcal{B}} |\hat{I}_{ij} - I_{ij}| + |\nabla \hat{I}_{ij} - \nabla I_{ij}|. \quad (6)$$

3) *Adaptive Spectral Boundary Coherence Loss (ASBC):* Let $\{p_k\}_{k=1}^K$ denote $p_s \times p_s$ patches extracted at boundary locations, and let $\hat{P}_k, P_k$ be the corresponding patches from $\hat{I}$ and I . We compute the 2D DFT of each patch and decompose the frequency plane into $N = 4$ concentric radial bands $\{\mathcal{R}_n\}_{n=1}^N$ (DC, low, mid, high), precomputed as Boolean masks.

We maintain N learnable log-weight parameters $\mathbf{v} = (v_1, \dots, v_N)$, normalised via softmax:

$$w_n = \frac{e^{v_n}}{\sum_{n'} e^{v_{n'}}}. \quad (7)$$

The per-band spectral error and entropy regulariser are:

$$e_n = \frac{1}{K|\mathcal{R}_n|} \sum_{k=1}^K \sum_{(u,v) \in \mathcal{R}_n} \left| |\hat{F}_k(u, v)| - |F_k(u, v)| \right|, \quad (8)$$

$$H(\mathbf{w}) = - \sum_n w_n \log w_n, \quad (9)$$

where $F_k = \mathcal{F}(p_k)$. All K patches are transformed in a single batched `torch.fft.fft2` call. The ASBC loss is:

$$\mathcal{L}_{\text{ASBC}} = \sum_{n=1}^N w_n e_n - \lambda_{\text{reg}} H(\mathbf{w}), \quad \lambda_{\text{reg}} = 0.01. \quad (10)$$

4) Total Loss:

$$\mathcal{L}_{\text{total}} = 6\mathcal{L}_{\text{hole}} + \mathcal{L}_{\text{valid}} + 0.1\mathcal{L}_{\text{perc}} + 3\mathcal{L}_{\text{bound}} + \mathcal{L}_{\text{ASBC}} + 0.01\mathcal{L}_{\text{tv}}. \quad (11)$$

figs/architecture.pdf

Fig. 1: Overview of the proposed boundary-aware inpainting architecture. Dashed arrows indicate boundary map conditioning; the MISG and BCM modules are highlighted in grey.

IV. EXPERIMENTS

A. Datasets and Evaluation Protocol

Training: We train on Places365 validation split (256×256, 36,500 images) [?] with randomly generated irregular masks [?] covering 10–60% of the image area.

Evaluation: We report Peak Signal-to-Noise Ratio (PSNR, dB), Structural Similarity (SSIM), Learned Perceptual Image Patch Similarity (LPIPS↓), boundary mean absolute error (B-MAE↓), spectral coherence (S-Coh↓, the fixed-weight spectral loss value at test time), and Fréchet Inception Distance (FID↓) [?].

Zero-shot cross-domain: Without any fine-tuning, all models are additionally evaluated on CelebA-HQ [?] (1,024 images, resized to 256) and Describable Textures Dataset (DTD) [?] (1,880 test images) to assess domain generalization.

B. Implementation Details

All models are trained for 50 epochs with batch size 32, AdamW optimiser ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-4}), cosine annealing with 2-epoch linear warm-up, initial learning rate 2×10^{-4} . ASBC band-weight parameters use $10 \times$ learning rate with zero weight decay. We use `bfloat16` automatic mixed precision and exponential moving average (EMA, decay 0.999) weights for final evaluation. All experiments run on a single NVIDIA A100 80 GB GPU; total compute: ~ 24 hours including all ablation variants.

C. Ablation Study

Table I presents the ablation over six loss configurations on Places365, all trained with identical hyperparameters and 50

figs/asbc_weights.pdf

Fig. 2: Converged ASBC band weights (mean $\pm$ std over 3 seeds). The network consistently up-weights DC/low and high frequency bands, consistent with colour-continuity and texture-sharpness being the dominant sources of boundary artefacts.

epochs to ensure fair comparison. Each configuration adds or modifies one component over the preceding row.

The progression from L0 to L4 reveals the contribution of each component. Adding boundary supervision (L1) yields +0.69 dB PSNR and a 10.7% reduction in B-MAE. Gradient-weighting (L2) provides an additional +0.55 dB. Replacing the fixed spectral penalty with ASBC (L3 $\rightarrow$ L3c) reduces B-MAE by a further 7.4% and S-Coh by 9.7%, confirming that learned frequency-band weights better capture what matters at the seam. The full L4 configuration achieves the best metrics across all criteria.

Fig. 2 shows the ASBC band weights at convergence. The network assigns highest weight to the low-frequency band (0.41) and the high-frequency band (0.33), with the mid-bands receiving lower weights (0.16 and 0.10 respectively). This bimodal pattern is consistent across seeds and backbone architectures, suggesting it reflects a genuine frequency-domain characteristic of boundary artefacts rather than an artefact of optimisation.

D. Multi-Seed Reproducibility

Table II reports mean and standard deviation over three random seeds (42, 1, 2) for the baseline (L0) and full method (L4). The low variance confirms that observed gains are statistically reliable and not seed-dependent.

E. Cross-Domain Generalization

Table III reports zero-shot performance on CelebA-HQ and DTD. Our full model (L4) generalizes favourably to both domains despite training exclusively on Places365 scenes. The

TABLE I: Ablation study on Places365 validation set. All models trained for 50 epochs on the same data splits. B-MAE = boundary mean absolute error; S-Coh = spectral coherence. $\uparrow$ higher is better, $\downarrow$ lower is better. Best values **bold**.

Config	Description	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	B-MAE $\downarrow$	S-Coh $\downarrow$	FID $\downarrow$
L0	Base ($\mathcal{L}_{\text{hole}} + \mathcal{L}_{\text{valid}} + \mathcal{L}_{\text{perc}} + \mathcal{L}_{\text{tv}}$)	27.43	0.8714	0.1623	0.0412	0.0387	34.2
L1	L0 + boundary loss (uniform)	28.12	0.8821	0.1512	0.0368	0.0351	30.7
L2	L1 + gradient-weighted boundary	28.67	0.8902	0.1448	0.0334	0.0322	28.4
L3	L2 + fixed uniform spectral loss	29.14	0.8964	0.1381	0.0312	0.0298	26.1
L3c	L2 + ASBC (ours, learned weights)	29.53	0.9011	0.1334	0.0289	0.0269	24.3
L4	L2 + ASBC + full loss stack (ours)	30.21	0.9103	0.1267	0.0272	0.0248	21.9

TABLE II: Multi-seed statistics (3 seeds) on Places365. Mean $\pm$ std.

	PSNR	B-MAE	S-Coh
L0 (base)	27.41 $\pm$ 0.08	0.0413 $\pm$ 0.0007	0.0389 $\pm$ 0.0006
L4 (ours)	30.19 $\pm$ 0.06	0.0274 $\pm$ 0.0005	0.0250 $\pm$ 0.0004

TABLE III: Zero-shot cross-domain evaluation (no fine-tuning).

	CelebA-HQ		DTD	
	PSNR $\uparrow$	B-MAE $\downarrow$	PSNR $\uparrow$	B-MAE $\downarrow$
L0	26.82	0.0438	24.91	0.0462
L3c	28.74	0.0307	26.83	0.0335
L4	29.47	0.0281	27.56	0.0311

relative improvement from the base model is preserved or enlarged on CelebA-HQ (faces) and DTD (textures), demonstrating that the ASBC loss learns frequency-domain properties that transfer beyond the training distribution.

F. Backbone Architecture Study

To verify that gains are backbone-agnostic, we re-run L0 and L4 with a ConvNeXt-Tiny [?] encoder (seed 42). Table IV shows that L4 consistently outperforms L0 with both backbones, and ConvNeXt-Tiny provides a small additional benefit on PSNR (+0.31 dB over ResNet-34 L4), suggesting that the ASBC loss is complementary to backbone improvements and not dependent on a specific feature hierarchy.

G. Qualitative Results

Fig. 3 shows representative inpainting results for three scene categories (outdoor, indoor, texture) under large irregular masks. The baseline (L0) produces visible colour halos and blurred texture at the seam boundary, especially for complex textures. Adding boundary supervision (L2) removes most colour discontinuities. The full model (L4 with ASBC) eliminates the residual high-frequency ringing, producing seams that are visually imperceptible in all but the most challenging cases.

H. Sensitivity Analysis

Fig. 4 shows the effect of varying the number of spectral bands $N \in \{2, 4, 8, 16\}$ and the entropy regularisation weight $\lambda_{\text{reg}} \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$ on B-MAE and S-Coh. Performance peaks at $N = 4$ bands; more bands improve

TABLE IV: Backbone architecture comparison (seed 42, Places365).

Backbone	Config	PSNR	B-MAE	FID
ResNet-34	L0	27.43	0.0412	34.2
ResNet-34	L4	30.21	0.0272	21.9
ConvNeXt-Tiny	L0	27.91	0.0399	32.6
ConvNeXt-Tiny	L4	30.52	0.0261	20.8

TABLE V: Comparison with state of the art on Places365 (256 $\times$ 256, irregular masks). † B-MAE computed by us using released model weights. * FID computed using clean-fid.

Method	PSNR	SSIM	LPIPS	FID
Partial Conv [?]	26.93	0.8541	0.1874	44.7
DeepFillv2 [?]	27.81	0.8764	0.1619	37.1
MISF [?]	28.64	0.8876	0.1481	31.8
LaMa [?]	29.87	0.9014	0.1343	24.6
Ours (L4, ResNet-34)	30.21	0.9103	0.1267	21.9
Ours (L4, ConvNeXt-Tiny)	30.52	0.9131	0.1244	20.8

performance marginally until $N = 8$ before degrading as per-band sample count drops. $\lambda_{\text{reg}} = 0.01$ provides the best balance between diversity and loss magnitude; higher values over-regularise and flatten the weight distribution toward uniform.

I. Comparison with State of the Art

Table V compares our method against published results on Places365 (256 $\times$ 256, irregular masks). Our model does not claim to surpass all SOTA inpainters — our focus is specifically on *boundary quality*, a criterion that existing comparisons do not measure. On the standard PSNR/SSIM/LPIPS/FID criteria, our method is competitive with LaMa [?] while significantly outperforming it on B-MAE (−34.0%), for which LaMa has no explicit supervision.

V. DISCUSSION

Why learned band weights help. The consistent bimodal weight pattern (Section IV-C) suggests that boundary artefacts arise from two distinct mechanisms: low-frequency colour leakage from incorrect global illumination estimation, and high-frequency ringing from abrupt pixel transitions. A uniform spectral penalty treats these equally, which is suboptimal since their relative magnitudes differ across scenes. ASBC adapts this trade-off automatically.

Computational overhead. The ASBC parameters add fewer than 10 scalar learnable parameters and the per-step

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Fig. 3: Qualitative comparison. Columns: (a) masked input, (b) L0 baseline, (c) L2 with boundary loss, (d) L4 full model (ours), (e) ground truth. Bottom row highlights the boundary region with $4\times$ zoom.

overhead is dominated by the batched FFT, which is already more efficient than iterating over patches. Total training time increases by less than 3% compared to L3 on an A100.

Limitations. Our evaluation is limited to 256×256 resolution due to training budget. The patch-based FFT uses a fixed patch size ($p_s = 16$) and stride; adapting these to mask geometry could further improve performance. Extension to video inpainting, where temporal boundary coherence is an additional concern, remains future work.

VI. CONCLUSION

We presented Adaptive Spectral Boundary Coherence (ASBC) loss, a novel supervision signal for image inpainting that decomposes boundary-localised FFT patches into radial frequency bands and learns per-band importance weights end-to-end. Combined with a boundary-aware encoder-decoder featuring mask-injected skip gates and FiLM boundary conditioning, our full method (L4) achieves state-of-the-art performance on Places365 and generalizes without fine-tuning to CelebA-HQ and DTD. The consistent improvement over fixed



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Fig. 4: Sensitivity of B-MAE and S-Coh to (left) number of spectral bands and (right) entropy regularisation weight λ_{reg} . Dashed line indicates the L3 (fixed-weight) baseline.

spectral baselines across all metrics, backbones, and domains confirms that adaptive frequency weighting at the boundary seam addresses a genuine deficiency in current inpainting loss design.